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Supplementary information

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Charting brain morphology in international healthy and neurological populations

In the format provided by the authors and unedited

Supplementary Materials

Supplementary Methods

Data collection

At the time of 3D T1-weighted image (isotropic spatial voxel size ≤ 1mm) collection, we collected a total of 30,249 scans including 25,424 scans of healthy controls (HCs), 372 scans with mild cognitive impairment (MCI), 802 scans with Alzheimer's disease (AD), 1,418 scans with Parkinson's disease (PD), 999 scans with cerebral small vessel disease (CSVD), 742 scans with multiple sclerosis (MS) and 492 scans with aquaporin-4 antibody seropositivity neuromyelitis optica spectrum disorder (NMOSD).

Inclusion and exclusion criteria are as follows: HCs were defined as individuals with no history of clinically-diagnosed CNS disorders aged 0 to 100 years; clinically-diagnosed AD was defined as patients meeting the criteria of the National Institute on Aging and Alzheimer's Association clinical criteria[1]; clinically-diagnosed idiopathic PD was defined according to the clinical criteria for PD by the International Parkinson and Movement Disorder Society[2]; clinically-diagnosed CSVD was defined according to the STandards for ReportIng Vascular changes on nEuroimaging criteria[3]; and clinically-diagnosed MS was determined as relapsing-remitting MS according to the 2017 McDonald criteria with both myelin oligodendrocyte glycoprotein and aquaporin 4 antibody seronegative[4]; AQP4+ NMOSD was determined according to antibody status (AQP4 antibody detected by cell-based immunofluorescence assay [CBA])[5]. Exclusion criteria were: (1) a history of other neurological or neuropsychological diseases (e.g., stroke or tumor); (2) incomplete MRI acquisition, poor image quality or insufficient image segmentation. For the cross-sectional analysis, we also excluded the follow-up scans from the main analysis. According to the exclusion criteria 1,273 HCs, 17 MCI, 100 AD, 192 PD, 40 CSVD, 151 MS and 13 NMOSD were excluded from the normative model fitting and disease comparison analysis (details see **Extended Data Fig. 1**). Finally, 24,061 HCs and those with neurological diseases including 308 MCI, 604 AD, 1,226 PD, 864 CSVD, 519 MS and 411 NMOSD participated in this study. The sample distribution across the 34 provinces was shown in **Extended Data Fig. 1**.

Data processing

T1-weighted images were processed and segmented using FreeSurfer Software Suite, version 7.3.2 (Laboratory for Computational Neuroimaging at the Athinoula A. Martinos Center for Biomedical Imaging; additional segmentation using FreeSurfer 6.0.1 was performed for comparisons). Briefly, the first processing stage of recon-all includes: non-uniformity correction, projection to Talairach space, intensity normalization, skull-stripping, automatic tissue and subcortical segmentation. Subsequently, surface interpolation, tessellation and registration are done at the second and third stages of the recon-all pipeline. All six brain tissue volumes were extracted from the aseg.stats files output by the recon-all process: "Total cortical gray matter volume" for GMV; "Total cerebral white matter volume" for WMV; "Subcortical gray matter volume" for sGMV (inclusive of thalamus, caudate nucleus, putamen, pallidum, hippocampus, amygdala, and nucleus accumbens area); the difference between "BrainSegVol" and "BrainSegVolNotVent" for Ventricular volume (termed CSF in this study); "Total cerebellar gray and white matter volume" for cerebellum volume; "Total brain-stem volume" for brainstem volume. Regional volume, cortical thickness and surface area were estimated for each of 34 cortical regions defined by the Desikan-Killiany parcellation atlas following the final stages of the recon-all pipeline and using the hemisphere-specific aparc.stats files generated by FreeSurfer. In addition, the mean cortical thickness and total surface area at the whole cortical level were also extracted. Further, the cortical measures (volume, thickness and area) based on additional atlases including HCP-MMP1.0 parcellation (Human Connectome Project Multi-Modal Parcellation version 1.0), DU15NET and Brainnetome atlases[6-8] were also extracted to extend the application of the Chinese normative references.

Supplementary Results

Supplementary Results 1. Comparisons of the brain structure measures and normative references derived from different FreeSurfer versions

To explore the potential contributions of the FreeSurfer versions on brain development

disparities between Chinese (CH) and European and North American (ENA) normative references, we conducted additional analyses to explore whether the disparities were related to FreeSurfer versions (here we compared version 6 [6.0.1] and version 7 [7.3.2], as we used version 7 in the CH normative references, while version 6 was adopted in the ENA normative references). We used the Concordance Correlation Coefficient (CCC) to assess the consistency between the raw measures and normative references (median curves) derived from the different FreeSurfer versions for the CH population. CCC is a statistical measure used to assess the agreement between two continuous variables. CCC incorporates both precision (correlation) and accuracy (bias correction), making it ideal for method comparison studies. The results showed high consistencies of the brain structural raw measures derived from the FreeSurfer version 7 and version 6 segmentations (CCC ranged from 0.91 to 1 for global raw measures). The normative references also showed high consistencies between FreeSurfer version 7 and version 6. Last, we observed similar milestones (peak ages) of the normative references (peak age differences ranged from 0.3 to 1.7 years, which were smaller than the differences of sGMV [4 years] and WMV [8.9 years] between CH and ENA normative references). Our findings confirmed that the brain development disparities between CH and ENA populations may be less dependent on the FreeSurfer segmentation algorithms (version 6 versus version 7, **Extended Data Fig. 2**).

Supplementary Results 2. Comparisons between CH and ENA normative references

To further investigate the disparities between CH and the ENA normative references. We used CCC to explore the similarity and difference between these normative references (median curves). Here we compared the normative references on both global and regional MRI measures. However, for regional brain structural measures, the ENA references only released normative references integrating the measures of bilateral areas as one region, which differ from our current references regarding a unilateral area as one region. In addition, in our CH normative references of regional brain structural measures, we found asymmetry for the bilateral hemispheres. Therefore, for the

normative reference comparisons between our fitted CH and ENA normative references, we compared the CH unilateral findings with the integrated ENA bilateral findings, separately. First, we observed that the ENA references seemed to fit our data well, except for cortical thickness displaying a large shift from CH data distribution, even after we calibrated the ENA references for site effects (calibrated by brain structural measures derived from FreeSurfer 6). The calibrated ENA normative references are still insufficient to fit our data, especially for cortical thickness. Large differences in median curves of cortical thickness were observed between CH normative references, and the raw and calibrated ENA normative references, as indicated by the low CCCs. These findings indicated that population-specific normative references are necessary especially for the distinct populations. Even though the calibrated ENA references seemed to provide a better fitting for different populations with site-calibration, the disparities still persisted, implying the importance of using population-specific normative references for clinical and research applications.

Supplementary Results 3. Comparisons of Chinese normative references using different fitting algorithms

To enhance the fitting confidence and the homogenization of data from multi-sites of the CH normative references by GAMLSS, we compared the constructed CH normative references with other popular algorithms. Here we used the hierarchical Bayesian regression framework as a comparative algorithm, as it was proved to have advantages in mitigating multi-site variability from multi-sites[9]. Here, we outlined a rigorous comparison between the GAMLSS and HBR farmwork.

In the main findings, we used the random effects in GAMLSS with generalized gamma distribution, same as the selection of Bethlehem et al. to account for the multi-site variability[10]. Consistent with the paper by Bethlehem et al.[10], we used the fractional polynomials, which can be viewed as a simpler form of spline modelling using a fixed set of polynomials. Prior to the GAMLSS fitting, we used a parameter optimization strategy similar with the Bethlehem et al.[10] to determine the optimal number of terms

in the fractional polynomial model, and whether to include the random effects (site) by selecting the minimal Bayesian information criterion (BIC) among multiple possible models. For comparisons with the HBR methods, we conducted a series of experiments regarding model fitting on global structural measures using HBR with normal distribution and a reparameterization of SHASH distribution, referring to the study of Augustijn et al.[9]. We compared the HBR-normal with normal likelihood, HBR-Sb1 with SHASHb likelihood (both ε and δ assume constant values), and HBR Sb2 (both ε and δ are linear regression on the design matrix, consistent with the paper by Augustijn et al. [9], it contains 5-knot b-spine basis expansion). These parameter settings were referenced to the study of Augustijn et al. [9].

First, we compared the fitting models on global brain structural measures (**Extended Data Fig. 3**). We observed the HBR model with normal and SHASH likelihoods could fit the data well, while the GAMLSS was more flexible for the local data tuning.

Second, we adopted two methods to assess the multi-site effect correction between GAMLSS and HBR (**Extended Data Fig. 3**). In the normative model fitting, if the site-effects could be well mitigated, the Z-scores for HCs in each site should be around zero. Therefore, we compared the Z-scores distribution of HCs across the sites for GAMLSS and HBR-Sb2 (which is proved to be the best to solve the site effects in the study of Augustijn et al.)[9]. We observed that the GAMLSS could well mitigate the site effects for most of the global measures when the random effect is considered in the GAMLSS model. While for WMV and CSF curve fitting, the random effects were not considered in the model, the Z-scores of HCs for some site may deviate from zero. For HBR-Sb2 method, we included the site as random effects in all models, the site effects were well mitigated, making the Z-scores around zero across sites.

In addition, we adopted the classification measures (area under the curve, AUC) as in the study of Augustijn et al.[9] to assess if the site variability was well mitigated. If the

site effects were mitigated, the classification AUCs of the Z-scores between each pair of sites should be around 0.5. We adopted a heatmap to display the AUC maps between each pair of sites for each global measure. Both GAMLSS and HBR-Sb2 showed AUCs around 0.5, indicating well mitigation for site variability. However, the WMV and CSF by GAMLSS showed relatively high AUCs between some sites, as these models did not include the random effects in the optimal possible model selection by BIC. For other measures, the AUC maps were comparable between GAMLSS and HBR-Sb2.

Considering the GAMLSS is a WHO recommended methods for normative reference fitting[11], and its higher fitting efficiency compared to HBR (for our data, GAMLSS: about 20 mins for each measure fitting [including the best model selection] versus HBR: about 2-3 hours for each measure fitting; using a computer with Linux system of Ubuntu 20.04.6, CPU of 2× Intel(R) Xeon(R) E5-2678 v3 @ 2.50GHz, and 24 physical cores, RAM of 62 GB, GPU of NVIDIA GeForce RTX 3090, graphic memory of 24 GB). Meanwhile, we also provided the normative references fitted by HBR-normal, HBR-Sb1 and HBR-Sb2 for extending applications, as HBR seemed to have a superior ability to account for the multi-site variability. We have uploaded the HBR models to the Science Data Bank (https://doi.org/10.57760/sciencedb.o00133.00065) for further external validation and application.

Supplementary Results 4. Resampling and comparisons of Chinese normative references using age censoring samples

To test the sensitivity of the estimated peak ages, particularly for samples from cases below 18 years, we performed additional sensitivity analyses by varying the sample size of cases aged 4-18 years. We down-sampled the pediatric data (4-18 years) at different ratios (from 0.1 to 1 in increments of 0.1, corresponding to sample sizes ranging from 208 to 2084). Each down-sampled subset was then combined with the original adult data (without resampling), and we refitted the curves across the full age range (4-85 years) while calculating bootstrap CIs (1000 bootstraps). This approach

allowed us to evaluate how the estimated peak ages varied with sample size and to determine the potential minimum pediatric sample size required for robust peak age estimation. As shown in **Extended Data Fig. 4b**, we applied this analysis to GMV, sGMV, WMV, CSF, cerebellum volume, brainstem volume, cortical thickness, and surface area. The fitted curves exhibited substantial variability at smaller pediatric sample sizes, particularly for cerebellum volume and surface area. However, as the sample size increased, the curves converged and stabilized. Similarly, the estimated peak ages for cerebellum volume and total surface area showed marked fluctuations when the pediatric sample size was small (less than 1000), but stabilized at larger sample sizes (e.g., above 1000). These results indicate that pediatric sample size significantly impacts peak age estimation, especially for cerebellum volume and surface area. In this study, the inclusion of 2084 scans from cases below 18 years ensures robust estimates of peak ages in this age group.

To enhance the robustness of the CH normative references and related findings, we fitted the normative curves for samples aged from 18 to 70 years, which showed narrow fitting confidence intervals (CIs) within such age ranges. Then we used the samples with ages of 18-70 years to construct the CH normative references and applied age censoring references to downstream clinical tasks including deviation summary across diseases, disease-specific classification and disease propensity scores, cognitive and physical ability prediction and prognostic tasks on PD and neuroinflammatory diseases. We have provided all these findings in **Extended Data Fig. 4**. The adjusted R-squared values of the age censoring fitting for global measures were 0.28-0.58, compared to those (also 0.28-0.58) using the samples with age ranging from 4 to 85 years. Their 1000 stratified bootstrap CI files were uploaded to GitHub (https://github.com/zhuozhizheng/Charting-Chinese-brain-health-and-neurological-disorders-across-lifespan) and Science Data Bank (DOI:10.57760/sciencedb.o00133.00065). Available peak ages were presented for WMV (36.9 [95% bootstrap CI 36.2-37.5]) and brainstem (34.3 [95% bootstrap CI 32.8-35.4]), similar to those using the samples with ages of 4-85 years. The

performances of the clinical tasks are comparable to the main findings using the normative references constructed by samples with age ranging from 4 to 85 years. Considering the generalization of the normative references, and comprehensive comparison to the ENA references, we kept the findings of 4-85 years in the main text, and provided these age censoring findings in the Supplementary materials for extending comparisons.

Supplementary Results 5. Normative references and clinical tasks stratified by sex

We observed that male subjects have larger gray matter and white matter volumes, larger ventricle volume and total surface area than female subjects (**Fig. 2**). The statistical analyses were consistent with the main text. The deviation score distributions in male and female patients are similar for these diseases except for MCI, MS and NMOSD that showed severe brain atrophy in female patients, indicating a different pathology-sensitivity of male and female patients in these diseases (**Extended Data Fig. 5a,b**).

The group differences of deviation scores were further evaluated using Cohen's d via Monte Carlo Test. This comparison was conducted between disease groups (comprising 3,932 patients) and age- and site-matched (if available; otherwise, unmatched) HCs. To emphasize the most sensitive deviation scores associated with disease pathology, we highlighted the deviation scores exhibiting the largest Cohen's d between HCs and each disease group. Furthermore, to assess the dissimilarity of the deviation score distribution of the most sensitive measures between diseases and HCs, we additionally calculated the overlapping coefficient, which was defined as the overlapped percentage of the distribution of deviation scores between groups. To estimate the univariate predictive ability of the most sensitive deviation scores for clinical diagnosis, we conducted univariate logistic regression with 10-fold cross validation for the most sensitive deviation scores, which was evaluated by accuracy and AUC.

We also observed distinct global and local deviation scores that displayed largest

difference between diseases and HCs for female and male patients especially for NMOSD. As NMOSD is a female-predominant disease, the small sample size of male patients resulted in a larger confidence interval for Cohen's d and overlapping coefficient. These sex-related differences may support previous findings on the sex-specific alterations in patients. These also provide clues about which diseases show a sex-specific alteration and which do not. These findings may be important for clinical management of patients according to the sex-specific pathology.

Supplementary Results 6. Comparison of clinical tasks using centile score, Z-score and raw measures

To demonstrate advantages of deviation scores derived from normative references, we compared the performances of clinical tasks among centile scores (we used in this study), Z-scores (the other normative reference derived deviation score) and raw measures (commonly used in conventional case-control studies). The results showed that both the Z-scores and raw measures of diseases displayed a normal distribution which is different from the centiles displaying a non-normal distribution. The statistical analyses were the same as those of centile scores in the main findings. We observed comparable Cohen's d, overlapping coefficient and classification performance among centile scores, Z-scores and raw measures. However, different regional deviation scores with the largest Cohen's d were observed in PD, MS and NMOSD, indicating the detection of pathology-related structural MRI measures may be linked with the adopted quantitative methods. However, we currently cannot determine whether the deviation scores are superior to raw measures in the detection of disease pathology, as previous findings were mainly based on raw measures in classical case-control designs. Additional analyses showed that these methods have comparable abilities for the detection of the number of features that showed statistical significance between disease groups and HCs. More than 50% of detected features were overlapped, except for NMOSD (only 36%), implying a combination of these methods may provide a robust detection of structural alterations in neurological diseases.

Supplementary Results 7. Additional univariate and multivariate analyses between disease groups, and longitudinal analyses

We provided additional findings on univariate and multivariate analyses to enhance the interpretation of the main findings.

Extended Data Fig. 9a provided effect sizes (Cohen's d) of global and regional structural measures across diseases. For details of the effect size on regional deviation scores (**Extended Data Fig. 9a**), we mapped the Cohen's d onto the subcortical and cortical regions. Based on these maps, we observed distinct effect sizes in subcortical volume, cortical volume, cortical thickness and surface area. These distinct local alterations may provide indications about which region may be the first target or which region is the most sensitive to the disease-specific pathology. These findings may also provide individual maps by using the normative references that help clinicians assess the changes underlying individual brains, which would aid clinical decision and individualized therapy.

To emphasize the most sensitive deviation scores associated with disease pathology, we highlighted the deviation scores exhibiting the largest effect sizes (largest Cohen's d) between HCs and each disease group. Furthermore, to assess the dissimilarity of the deviation score distribution of the most sensitive measures between diseases and HCs, we additionally calculated the overlapping coefficient, which was defined as the overlapped percentage of the distribution of deviation scores between groups. To estimate the univariate predictive ability of the most sensitive deviation scores for clinical diagnosis, we conducted univariate logistic regression for the most sensitive deviation scores, which was evaluated by accuracy and AUC (**Extended Data Fig. 9b**).

For deviation scores in neurological diseases, we reported the global and regional deviation scores that displayed the largest effect sizes in each disease group compared with HCs. For global deviation scores, MCI (Cohen's d = 0.66 [95% CI 0.46-0.86], false discovery rate corrected p [pFDR] < 0.001), AD (Cohen's d = 0.82 [95% CI 0.69-

0.96], pFDR < 0.001), PD (Cohen's d = 0.60 [95% CI 0.53-0.68], pFDR< 0.001), CSVD (Cohen's d = 0.59 [95% CI 0.46-0.71], pFDR < 0.001), and NMOSD (Cohen's d = 0.33 [95% CI 0.19-0.48], pFDR < 0.001) showed the largest effect sizes in cortical thickness. MS (Cohen's d = 0.62 [95% CI 0.50-0.74], pFDR < 0.001) showed the largest effect size in sGMV. For regional deviation scores, MCI showed the largest effect size in left entorhinal volume (Cohen's d = 0.55 [95% CI 0.36-0.75], pFDR < 0.001). AD showed the largest effect size in left amygdala volume (Cohen's d = 1.06 [95% CI 0.93-1.20], pFDR < 0.001). PD showed the largest effect size in right fusiform thickness (Cohen's d = 0.68 [95% CI 0.60-0.76], pFDR < 0.001). CSVD showed the largest effect size in right thalamus volume (Cohen's d = 0.58 [95% CI 0.45-0.71], pFDR < 0.001). MS showed the largest effect size in right putamen volume (Cohen's d = 0.81 [95% CI 0.68-0.93], pFDR < 0.001). NMOSD showed the largest effect size in right fusiform volume (Cohen's d = 0.92 [95% CI 0.76-1.07], pFDR < 0.001) (**Extended Data Fig. 9b**).

We also calculated the effect size between disease pairs to investigate the difference and similarity between the deviation scores of diseases (**Extended Data Fig. 9c**). The statistical analyses were similar to those between diseases and HCs, as described in the main text. This may provide clues to clinicians about which diseases have distinct brain alterations and which diseases have similar brain changes even though distinct pathologies were found in these diseases. Regarding neurological diseases, some common pathologies are presented including the blood-brain barrier damages, oxidative stress, neuronal and axonal loss, which may pave a common target for these diseases. These findings would provide indications about which diseases may be managed with a common strategy.

The multivariate classification among neurological diseases were also conducted. We observed that the deviation scores could distinguish among different neurological diseases with high performances (AUCs ranged from 0.80 to 0.98), of which the classification of MCI versus AD and MS versus NMOSD showed relatively low AUCs of 0.82 and 0.80 respectively (**Extended Data Fig. 9d**).

We calculated the DPS using a one-vs-all strategy. That is focusing on the target disease (for each disease-specific classification model) and non-target cases (other diseases and HCs, these cases were age- and sex-matched). This one-vs-all strategy is more informative for differential diagnosis. For developing a disease-specific DPS model development, we used all other disease and HC samples for references (age- and sex-matched, but no site-matched, with consideration given to clinical generalizability and application in other unseen sites). Here, due to the relatively larger sample size in HCs, we first sampled the HCs with a ratio of 1:1 with the target disease as reference, and then combined the sampled HCs and other disease groups as a whole for the following age- and sex- matched samples. This step would help avoid an excessive number of HCs and balance the samples of HCs and other diseases. In addition, we provided the contributing measures selected in all the 10-fold cross-validation models in **Extended Data Fig. 9e** and the confusion matrix for each disease-specific classification models in **Extended Data Fig. 9f**. We also performed ANOVA to test the difference in estimated DPSs among disease groups for each disease-specific classification model, and found the disease-specific classification model had higher predicted probability for the target diseases compared to other diseases, indicating the high specificity of DPS for the target disease (**Extended Data Fig. 9g**).

Further, to enhance the robustness of the predictions of clinical scores, we provided the Pearson's correlation coefficient (r value) and MAPE distribution based on 1000 permuted brain structural deviation scores. These permuted distributions enhanced the robustness of the correlation and MAPE reported in the main text (**Extended Data Fig.9h**). We observed distinct distributions for R and MAPE, reflecting the different dimensions these metrics assess. The R value primarily quantifies the linear association between actual and predicted values, and its significance indicates that the predictions preserve the original covariation patterns in the data. In contrast, MAPE measures absolute prediction error at the individual level, reflecting the percentage errors in subject-specific predictions. Such divergence between R and MAPE may arise from

high-leverage points in the data that disproportionately influence the R value but contribute minimally to MAPE. Systematic biases in the model (e.g., consistent underestimation for high-value predictions) may also lead to the divergence between R and MAPE. Further, for a better interpretation of the associations between the deviation scores and clinical scores, we additionally calculated the univariate Pearson's correlations and ordered the deviation scores that survived the FDR correction across all variables and diseases (examples were shown in **Extended Data Fig.9k**). More findings were uploaded as **Supplementary Data-1**.

Finally, we provided the analyses of stability of deviation scores in HCs and clinical value of deviation score in monitoring disease development using follow-up MR scans. The median deviation scores for global and regional measures were all around 0 (ranging from -0.013 to 0.032) across follow-up time and ages (**Extended Data Fig. 10a**), indicating the stability of the deviation scores in HCs during the normal aging. To assess whether the deviation scores have a clinical value for monitoring the disease development. We adopted a linear mixed model that considered the diagnosis and follow-up time, as well as their interaction as fixed effects, and individuals as random effects (both intercept and slope of follow-up time were considered). Compared with those of the baseline scans (**Extended Data Fig. 10b**), the deviation scores of several structural measures (e.g., the GMV in MCI, and sGMV in AD, CSVD, MS and NMOSD patients) were lower for the follow-up scans ($p < 0.05$), but the differences did not survive FDR correction, with all pFDRs > 0.05.

Extended Data Figure legends

Extended Data Fig. 1 Details of included participants and their distributions. **a**, A brief flowchart of the included participants in this study. **b**, Sample distributions across HC and disease groups, provinces and ages (HC, n = 24061; diseases, n = 3932). **c**, The distributions of Euler Number across age, sex and diagnostic groups (n = 27993). Note: HC, healthy control; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum

disorder; yr, year.

Extended Data Fig. 2 Comparisons of normative references by different FreeSurfer versions, different populations, and extended brain atlases. **a**, The consistency analysis of the raw global and regional brain structural measures derived from FreeSurfer 6 and 7. **b**, The CH normative references fitted by the segmentation outputs of FreeSurfer 6 and 7, with the ENA and ENA-calibrated normative references as comparisons. The units are $\times 10^4 mm^3$ for volume measures, $\times 10^4 mm^2$ for area measure, and mm for thickness measure. **c**, consistency analysis of the normative references (median curves) between different FreeSurfer version and populations. **d,** The milestones (peak ages) of normative references constructed by outputs of FreeSurfer 6. **e,** The peak age differences of regional structural normative references. Only the available peak ages in both CH and ENA normative references were displayed. **f,** The peak ages of the normative references based on the extended brain parcellation atlas, including HCP-MMP1.0 parcellation (Human Connectome Project Multi-Modal Parcellation version 1.0), DU15NET, and Brainnetome Atlases. Note: CH, Chinese; ENA, European and North American; CCC, Concordance Correlation Coefficient; GMV, gray matter volume, sGMV, subcortical gray matter volume; WMV, white matter volume; CSF, cerebrospinal fluid; v6, FreeSurfer version 6; v7, FreeSurfer version 7; yr, year.

Extended Data Fig. 3 Comparisons of normative references by GAMLSS and HBR framework. **a,** The normative references fitted by GAMLSS and HBR framework. The units are $\times 10^4 mm^3$ for volume measures, $\times 10^4 mm^2$ for area measure, and mm for thickness measure. **b,** The Z-scores across the site in GAMLSS and HBR fitting (n = 105 sites), the closer the median value to zero, the better the site effects mitigated. **c,** The classification AUC maps of each paired sites (n = 105 sites). The closer the AUCs to zero, the better the site effects mitigated. Note: GMV, gray matter volume, sGMV, subcortical gray matter volume; WMV, white matter volume; CSF, cerebrospinal fluid; GAMLSS, generalized additive models for location, scale,

and shape; HBR, hierarchical Bayesian regression; AUC, area under the curve.

Extended Data Fig. 4 Normative references by age censoring samples (aged 18-70 years) and downstream clinical tasks. **a,** The fitted median curves and associated bootstrap CIs, of which, a majority of these global structural measures have narrow CIs, indicating a robust model fitting. The units are $\times 10^4 mm^3$ for volume measures, $\times 10^4 mm^2$ for area measure, and mm for thickness measure. **b,** Sensitivity analysis of the estimated peak ages and median fitting curves with different resampled count of cases below 18 years. The units are $\times 10^4 mm^3$ for volume measures, $\times 10^4 mm^2$ for area measure, and mm for thickness measure. **c,** The normative references constructed by age censoring samples (age limited to 18-70 years, n = 21,205). The units are $\times 10^4 mm^3$ for volume measures, $\times 10^4 mm^2$ for area measure, and mm for thickness measure. **d,** Deviation score summary across diseases and global measures. **e,** Deviation score summary across diseases and regional measures. **f,** Milestones (peak ages) of age censoring normative references. **g-j**, Downstream clinical tasks using the age censoring normative references. Note: AUC, area under the curve; DPS, disease propensity score; HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum disorder; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; BVMT, Brief Visuospatial Memory Test; CVLT, California Verbal Learning Test; SDMT, Symbol Digit Modalities Test; PASAT, Paced Auditory Serial Addition Test; UPDRS, Unified Parkinson's Disease Rating Scale; EDSS, Expanded Disability Status Scale; MAPE, mean absolute percentage error; DBS, deep brain stimulation; SPMS, secondary progressive multiple sclerosis; FDR, false discovery rate. All statistical tests were two-sided with FDR correction for multiple comparisons (if relevant). * pFDR < 0.05, ** pFDR < 0.005, *** pFDR < 0.001.

Extended Data Fig. 5 Deviation scores of global and regional structural measures in neurological diseases for female and male cases. **a,** Deviation scores of global

structural measures in neurological diseases for female and male cases. Statistical significances were presented for each disease group compared to age- sex- and site-matched HCs. * pFDR<0.05, ** pFDR<0.005, *** pFDR<0.001. **b,** Deviation scores of regional structural measures in neurological diseases for female and male cases. **c,** A summary of the most sensitive global and regional deviation scores (largest Cohen's d) in each neurological disease, together with the corresponding overlapping coefficient and classification metrics. Note: HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum disorder; ACC, accuracy; AUC, area under the curve; FDR, false discovery rate. All statistical tests were two-sided with FDR correction for multiple comparisons (if relevant).

Extended Data Fig. 6 Comparisons of clinical utility for detecting the alterations of global and regional brain structural measures in disease groups using Z-score and raw measures in conventional case-control methods. **a,** The Z-score and raw measure of global measures across diseases. * pFDR<0.05, ** pFDR<0.005, *** pFDR<0.001. **b,** The Z-score and raw measure of regional measures across diseases. For raw measures, the t-values between disease and HCs were mapped on to the subcortical and cortical areas. **c,** A summary of the most sensitive global and regional deviation scores (largest Cohen's d) in each neurological disease, together with the corresponding overlapping coefficient and classification metrics. **d,** the overlapped brain structural measures that showed statistical significance among the centile score, Z-score and raw measure. Note: GMV, gray matter volume, sGMV, subcortical gray matter volume; WMV, white matter volume; CSF, cerebrospinal fluid; HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum disorder; ACC, accuracy; AUC, area under the curve; FDR, false discovery rate. All statistical tests were two-sided with FDR correction for multiple comparisons (if relevant).

Extended Data Fig. 7 Comparisons of clinical utility of deviation score for DPS estimation using centile score, Z-score and raw measure. a, Disease-specific model performance for classification of disease and HCs, and predicted DPS for HCs and diseases using different disease-specific models. **b,** Distribution of the predicted DPS for targeted disease and non-target groups. The disease-specific model should have high DPS for targeted diseases and low DPS for other groups. The targeted diseases using the corresponding disease-specific models were highlighted. The statistical test was conducted by ANOVA among centile score, Z-score and raw measure for each disease group. Note: AUC, area under the curve; DPS, disease propensity score; HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum disorder.

Extended Data Fig. 8 Comparisons of clinical utility for the prediction of clinical variables for HCs and diseases involving cognitive and physical scores using centile score, Z-score and raw measure. **a,** The Pearson's correlation between predicted and actual clinical measures and the MAPE of predictive models were presented. * pFDR < 0.05, ** pFDR < 0.005, *** pFDR < 0.001. **b,** The predicted PD motor outcomes by multivariate deviation scores. **c,** The risk-stratified MS and NMOSD by multivariate deviation scores. Note: HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum disorder; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; BVMT, Brief Visuospatial Memory Test; CVLT, California Verbal Learning Test; SDMT, Symbol Digit Modalities Test; PASAT, Paced Auditory Serial Addition Test; UPDRS, Unified Parkinson's Disease Rating Scale; EDSS, expanded disability status scale; MAPE, mean absolute percentage error; DBS, deep brain stimulation; SPMS, secondary progressive multiple sclerosis; FDR, false discovery rate. All statistical tests were two-sided with FDR correction for multiple comparisons (if relevant).

Extended Data Fig. 9 Additional univariate and multivariate analyses between disease groups. **a,** The univariate Cohen's d distribution of global and regional brain structural measures across diseases groups. **b,** A summary of the global and regional deviation scores showing largest Cohen's d and corresponding overlapping coefficient and classification performance between disease groups and HCs. **c,** The univariate Cohen's d between each paired disease groups. * pFDR<0.05, ** pFDR<0.005, *** pFDR<0.001. **d,** Multivariate classification between disease groups. SVM with 10-fold cross-validation was conducted, similar to the DPS model development. **e,** The contributing measures of the disease-specific classification models for DPS estimation. For each disease, the measures that selected in all the SVM 10-folds were defined as contributing measures, and they were ordered by their univariate Cohen's d (HCs versus disease) in a decreasing manner. **f,** The confusion matrix of the disease-specific classification models for each disease. **g,** The distribution of DPS across diseases by using disease-specific classification models. ANOVA was used for statistical analysis. **h,** The distributions of 1000 permuted Pearson's correlation and MAPE for clinical score and PD outcome prediction. **j,** The PD outcome prediction using baseline EPDRS alone. **k,** The univariate Peason's correlation between the deviation scores clinical scores across diseases. Note: MCI, mild cognitive impairment; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum disorder; AUC, area under the curve; GMV, gray matter volume; sGMV, subcortical gray matter volume; WMV, white matter volume; CSF, cerebrospinal fluid; SVM, support vector machine; DPS, disease propensity score; MAPE, mean absolute percentage error; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; UPDRS, Unified Parkinson's Disease Rating Scale; EDSS, Expanded Disability Status Scale; DBS, deep brain stimulation; FDR, false discovery rate. All statistical tests were two-sided with FDR correction for multiple comparisons (if relevant).

Extended Data Fig. 10 Assessment of the stability of deviation scores and disease

development using longitudinal MR scans. **a,** The deviation score changes of follow-up scans across measures and follow-up times in HCs. **b,** The deviation score changes of follow-up scans across measures and disease in a linear mixed model. **c**, An actionable clinical use of individual brain health report. A case of individual DPS and associated brain structural measure deviations. This case indicates that the DPS estimated by centile score is superior to that by raw measure to aid clinical decision-making. Note: HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer's disease; PD, Parkinson's disease; CSVD, cerebral small vessel disease; MS, multiple sclerosis; NMOSD, neuromyelitis optica spectrum disorder; m, month; yr, year; GMV, gray matter volume; sGMV, subcortical gray matter volume; lh, left hemisphere; rh, right hemisphere; DPS, disease propensity score; CI, confidence interval. All statistical tests were two-sided.

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